

Grade 8
Unit 4
Lesson 7 Practice

Name Answer key
Date _____
Period _____

Apply the laws of exponents to generate an equivalent algebraic expression in questions 1-6.

1. $x^7 \cdot x^4$ x^{11}

2. $7y \cdot 6y^{10}$ $42y^{11}$

3. $(c^3)^5$ c^{15}

4. $(8z^2)^3$ $512z^6$

5. $4a^4b^6 \cdot 2a^3b^2$ $8a^7b^8$

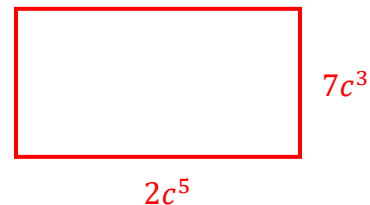
6. $(d^5 \cdot 6c^3)^2$ $36c^6d^{10}$

7. Match the algebraic expressions from Column A to an equivalent algebraic expression in Column B.

Column A	Column B
A. $(xy)^0 \cdot (5x^3 \cdot 2x^7)$ $1 \cdot 10x^{7+3}$	E. $7x^7y^7$
B. $((xy)^2)^3 \cdot 7xy$ $(xy)^6 \cdot 7xy$ $x^6y^6 \cdot 7xy$	F. $36x^{14}$
C. $(6x^7)^2$ $36x^{7 \cdot 2}$	G. $10x^{10}$
D. $(x^2y)^4 \cdot (4x)^3$ $x^8y^4 \cdot 4^3x^3$	H. $64x^{11}y^4$

A rectangle has a length represented by the expression $2c^5$ and a width represented by the expression $7c^3$.

8. Draw a rectangle and label the dimensions.



9. The formula $\text{Area} = \text{length} \times \text{width}$ can be used to find the area of the rectangle. Using the formula, generate an algebraic expression to represent the area of the rectangle.

$$2c^5 \cdot 7c^3$$

10. Simplify the expression from question 9 to find an equivalent expression that represents the area of the rectangle.

$$\text{Area} = 14c^8$$